Class 14

Domination by mixed strategies

	L	C	R
T	6,2	0,6	4,4
M	2,12	4,3	2,5
B	0,6	10,0	2,2

$\sigma_1 = \left[\frac{1}{2}(T), \frac{1}{2}(B)\right]$

M

$$U_1(\sigma_1, s_2) > U_1(M, s_2) \quad \forall s_2 \in S_2$$

$$\Rightarrow U_1(\sigma_1, \sigma_2) > U_1(M, \sigma_2) \quad \forall \sigma_2 \in \Sigma_2$$

A strategy σ_i of player i strictly dominates a pure strategy s_i of player i if

$$U_i(\sigma_i, \sigma_{-i}) > U_i(s_i, \sigma_{-i}) \quad \forall \sigma_{-i} \in \Sigma_{-i}$$

$$\Leftrightarrow U_i(\sigma_i, s_{-i}) > U_i(s_i, s_{-i}) \quad \forall s_{-i} \in S_{-i}$$

	L	C	R
T	6,2	0,6	4,4
B	0,6	10,0	2,2

$\left[\frac{5}{12}(L), \frac{7}{12}(C)\right]$ strictly dominates R. for player 2

	L	C
T	6,2	0,6
B	0,6	10,0

Theorem Suppose for player i, a strategy σ_i strictly dominates a pure strategy s_i of player i.

Then s_i does not get a positive probability in all NE of the game.

Proof Suppose $\bar{\sigma}_i$ is a strategy of player i s.t. $\bar{\sigma}_i(s_i) > 0$. We show that there is a mixed strategy $\bar{\bar{\sigma}}_i$ s.t. $\bar{\bar{\sigma}}_i$ strictly dominates $\bar{\sigma}_i$.

Construct $$\bar{\bar{\sigma}}_i(t_i) = \begin{cases} \bar{\sigma}_i(s_i)\cdot\sigma_i(s_i) & t_i = s_i \\ \bar{\sigma}_i(t_i) + \bar{\sigma}_i(s_i)\sigma_i(t_i) & t_i \neq s_i \end{cases}$$

$$\sum_{t_i \in S_i} \bar{\bar{\sigma}}_i(t_i) = \sum_{t_i \neq s_i} \left(\bar{\sigma}_i(t_i) + \bar{\sigma}_i(s_i)\,\sigma_i(t_i) \right) + \bar{\sigma}_i(s_i)\,\sigma_i(s_i)$$

$$= \sum_{t_i \neq s_i} \bar{\sigma}_i(t_i) + \sum_{t_i \in S_i} \left(\bar{\sigma}_i(s_i)\,\sigma_i(t_i) \right)$$

$$= \sum_{t_i \neq s_i} \bar{\sigma}_i(t_i) + \bar{\sigma}_i(s_i) \underbrace{\sum_{t_i \in S_i} \sigma_i(t_i)}_{=1}$$

$$= \sum_{t_i \neq s_i} \bar{\sigma}_i(t_i) + \bar{\sigma}_i(s_i)$$

$$= \sum \bar{\sigma}_i(t_i) = 1$$

$t_i \in S_i$

Fix $\sigma_{-i} \in \Sigma_{-i}$

$$u_i(\bar{\sigma}_i, \sigma_{-i}) = \sum_{t_i \in S_i} u_i(t_i, \sigma_{-i})\,\bar{\sigma}_i(t_i)$$

$$= \sum_{t_i \neq s_i} u_i(t_i, \sigma_{-i})\,\bar{\sigma}_i(t_i) + u_i(s_i, \sigma_{-i})\,\bar{\sigma}_i(s_i)$$

$$< \sum_{t_i \neq s_i} u_i(t_i, \sigma_{-i})\,\bar{\sigma}_i(t_i) + u_i(\sigma_i, \sigma_{-i})\,\bar{\sigma}_i(s_i) \quad \Big| \; u_i(s_i, \sigma_{-i}) < u_i(\sigma_i, \sigma_{-i})$$

$$= \sum_{t_i \neq s_i} u_i(t_i, \sigma_{-i})\,\bar{\sigma}_i(t_i) + \bar{\sigma}_i(s_i) \sum_{t_i \in S_i} u_i(t_i, \sigma_{-i})\,\sigma_i(t_i)$$

$$= \sum_{t_i \neq s_i} u_i(t_i, \sigma_{-i}) \left(\bar{\sigma}_i(t_i) + \bar{\sigma}_i(s_i)\,\sigma_i(t_i)\right)$$

$$+ u_i(s_i, \sigma_{-i})\,\bar{\sigma}_i(s_i)\,\sigma_i(s_i)$$

$$= \sum_{t_i \neq s_i} u_i(t_i, \sigma_{-i})\,\bar{\bar{\sigma}}_i(t_i) + u_i(s_i, \sigma_{-i})\,\bar{\bar{\sigma}}_i(s_i)$$

$$= \sum_{t_i \in S_i} u_i(t_i, \sigma_{-i})\,\bar{\bar{\sigma}}_i(t_i) = u_i(\bar{\bar{\sigma}}_i, \sigma_{-i}).$$

Proof of Nash's Theorem

Theorem Suppose G is a strategic form game where all the players have finitely many pure strategies. Then G has a mixed strategy NE.

Proof $G = \langle N, (S_i)_{i \in N}, (u_i)_{i \in N} \rangle$

$|S_i| = m_i \quad \forall \; i = 1(1)n$

Brouwer's fixed point theorem Let X be a convex and compact subset of $\mathbb{R}^d$ for some $d \in \mathbb{N}$ and let $f : X \to X$ be a continuous function. Then, f has a fixed point, i.e. $\exists \; x \in X$ s.t. $f(x) = x$.

Claim-1 $\Sigma_i = \left\{ (x_1, \ldots, x_{m_i}) \;\middle|\; x_j \geq 0 \;\; \forall j = 1(1)m_i, \; \sum_{j=1}^{m_i} x_j = 1 \right\}$ is a convex and compact subset of $\mathbb{R}^{m_i}$.

Claim-2 Suppose A and B are two convex and compact subsets of $\mathbb{R}^{d_1}$ and $\mathbb{R}^{d_2}$, respectively. Then $A \times B$ is a convex and compact subset of $\mathbb{R}^{d_1 + d_2}$.

From

Claim 1 and Claim 2 : $\Sigma = \Sigma_1 \times \Sigma_2 \times \cdots \times \Sigma_n$ is a convex and compact subset of $\mathbb{R}^{\sum_{i=1}^{n} m_i}$

We will construct a funct $f : \Sigma \to \Sigma$ s.t

(i) f is continuous

(ii) every fixed point of f is an NE of G.

$\sigma \in \Sigma$

σ is not NE, $\exists\ i \in N$ s.t. $\sigma_i \notin br_i(\sigma_{-i})$

$$g_i^j(\sigma) = \max\{0, U_i(s_i^j, \sigma_{-i}) - U_i(\sigma)\} \qquad \begin{matrix} i = 1(1)n \\ j = 1(1)m_i \end{matrix}$$

Claim-3 g_i^j is a cont function of σ $\forall i = 1(1)n$, $\forall j = 1(1)m_i$

Proof: $U_i(\sigma)$ is a multilinear function of σ, hence continuous.

$U_i(s_i^j, \sigma_{-i})$ is again a multilinear function in σ_{-i} and hence in σ, so, a cont function

$U_i(\sigma) - U_i(s_i^j, \sigma_{-i})$ is continuous

$\max\{0, U_i(\sigma) - U_i(s_i^j, \sigma_{-i})\}$ is also cont.